## Exam Instructions

- Make sure that exam paper include 22 pages.
- All MCQs have to be answered in bubble sheet.
- Write answer of MCQs in bubble sheet by blue pen or pencil.
- Double answer of any MCQ will be considered wrong answer.
- All easy questions have to be answered.

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## Multiple Choice Questions (MCQs)

**Choose the correct answer (60 MCQs): Total 30 marks (0.5 mark for each question)**

**Q1.** What are the common pathogens present in chronic obstructive pulmonary disease (COPD) patient?

a. Streptococcus pneumonia ,Haemophilus influenza
b. Staphylococcus pneumonia, Haemophilus influenza
c. Streptococcus pneumonia, acid fast bacillus
d. Klepsiella pneumonia

**Q2.** Haemoptysis and upper lobe pneumonia occur in:

a. Chlamydia pneumonia
b. Legionella pneumonia
c. Friedlander pneumonia
d. Community acquired pneumonia (CAP)

**Q3.** Hemochromatosis is most commonly associated with which of the following cardiomyopathies?

a. Congestive
b. Dilated
c. Restrictive
d. Hypertrophic

**Q4.** In treatment of cardiac failure associated with acute pulmonary edema:

a. Controlled oxygen therapy should be restricted to 28% oxygen in patient who smoke
b. Furosemide therapy given intravenously reduces preload & after-load
c. Nitrates should be avoided if systolic blood pressure < 140 mmHg

d. Angiotensin converting enzyme inhibitors decrease the after-load but increase the preload

Q5. A white milky appearance on the pleurocentesis would suggest a:

a. Thoracic duct injury
b. Methothelioma
c. Squamous cell carcinoma
d. Parapneumonic effusion

Q6. Which of the following is a common presenting symptom of bronchogenic carcinoma?

a. Wheezing
b. Hypercalcemia
c. Pancytopenia
d. Gynecomastia

Q7. A 30-year-old patient with a history of mild persistent asthma (baseline peak expiratory flow rate of 85%) presents to the emergency department with shortness of breath and wheezing that has not relieved by her albuterol inhaler for the past 12 hours. She was able to tolerate pulmonary function tests and a set was performed. Which of the following is the most likely test result?

a. Decreased $FEV_1$, normal/increased FVC, decreased $FEV_1$: FVC ratio, with post bronchodilator
b. $FEV_1$ increased by 13%.
c. Decreased residual volume and total lung capacity.
d. Increased $FEV_1$ increased FVC, normal $FEV_1$: FVC ratio.

Q8. What drug therapy should be available to ALL asthma patients (intermittent and persistent)?

a. Low-dose inhaled corticosteroids (ICS).
b. Short-acting Beta-agonist (SABA) on demand (PRN).
c. Long-acting Beta-agonist (LABA).
d. Leukotriene Antagonist.

Q9. The hallmark of asthma that distinguishes it from other obstructive airway diseases is that in asthma?

a. Hyperinflation is present on chest roentgenogram.
b. Airway obstruction is reversible.

c. Hypoxia occurs as a consequence of ventilation-perfusion mismatch.
d. The FEV1/FVC ratio is reduced.

**Q10.** Which of the following is the most accurate investigation to diagnose bronchiectasis?
a. Chest x ray.
b. High resolution computed tomography (HRCT).
c. Pulmonary function tests.
d. Arterial blood gases.

**Q11.** Beside bronchiectasis, which of the following manifestations is typical of Kartagener's Syndrome?
a. Intestinal obstruction.
b. Dextrocardia.
c. Steatorrhea.
d. Interstitial pulmonary fibrosis.

+ chronic sinusitis
+ situs inversus

**Q12.** Lung abscess is a type of ..............................
a. liquefactive necrosis.
b. Coagulative necrosis.
c. Fat necrosis.
d. Caseous necrosis.

**Q13.** Which of the following is a common predisposing factor for developing lung abscess?
a. Chronic obstructive air way disease (COPD)
b. Pulmonary embolism
c. Aspiration pneumonia
d. Bronchiectasis.

**Q14.** A 66 years old lady presented to emergency room (ER) with sudden acute chest pain and shortness of breath. Her blood pressure was 45\30 mmHg, pulse 132, respiratory rate 35, Oxygen saturation ($SpO_2$) was 81%. There was redness and hotness in left lower limb. In this case the heparin therapy for pulmonary embolism must be started:
a. On clinical suspicion
b. After confirmation of diagnosis
c. After confirmation of right ventricular (RV) dysfunction
d. After confirmation of deep venous thrombosis (DVT)

Q15. A firefighter who was involved in extinguishing a house fire is being treated for smoke inhalation. He develops severe hypoxia 48 hours after the incident, requiring intubation and mechanical ventilation. Which of the following conditions has he most likely developed?

a. Acute respiratory distress syndrome (ARDS).
b. Atelectasis.
c. Bronchitis.
d. Pneumonia.

Q16. Second line anti tuberculous drugs are indicated when?

a. Drug resistant organisms are known or suspected.
b. Polymerase chain reaction (PCR) is positive.
c. Computed Tomography (CT) scan shows extensive lesion.
d. Patients aged 70 years and more.

Q17. What is correct statement about malignant pleural effusion?

a. Milky effusion
b. Low pH and low glucose
c. Raised pleural fluid cholesterol
d. The presence of pus

Q18. Observation in pneumothorax required in ....................

a. Large pneumothorax with minimal symptoms
b. Tension pneumothorax
c. Large pneumothorax with severe symptoms
d. Small pneumothorax with no symptoms

Q19. Which of the following signs or symptoms is not found in tension pneumothorax?

a. Increased jugular venous pressure, cyanosis and dyspnea
b. Hypertension
c. Hyperresonance to percussion and absence of BS on the affected side
d. Deviation of the trachea to the contralateral side

Q20. Which of the following is the most accurate indication for mediastinoscopy?

a. Diagnosis of mediastinal lesion
b. Diagnosis of diaphragmatic hernia

c. Diagnosis of cardiac lesion
d. Diagnosis of vascular lesion

Q21. Hoarseness of voice is due to compression on:
a. Sympathetic trunk
b. Left recurrent laryngeal nerve
c. Esophagus
d. Trachea and bronchi

Q22. Which diagnostic method that differentiate between mesothelioma and Adenocarcinoma?
a. Pleural fluid chemical analysis
b. Computed tomography (CT) scan of chest
c. Chest x-ray
d. Immunohistochemistry

Q23. Which of the following jobs is associated with high risk to develop malignant mesotheliomas?
a. Bird breeders
b. Medical staff.
c. Farmers.
d. Insulation workers

Q24. Which of the following used in differentiation of acute from chronic respiratory failure?
a. Chest wheezes.
b. Pulmonary hypertension.
c. Cyanosis.
d. Anemia.

Q25. A 35-year-old man is brought to the Emergency Room (ER) unresponsive by ambulance. He is cyanotic with a BP of 100/80 and pulse of 75. His arterial blood gases on room air are $PaO_2$= 45 mmHg; $PaCO_2$= 75 mmHg; pH= 7.12. What is the most likely diagnosis for the blood gases are:
a. Metabolic acidosis
b. Hypoxic respiratory failure
c. Hypercapnic respiratory failure
d. Combined respiratory and metabolic acidosis

Q26. Which of the following is an accepted indication of acute oxygen therapy?

a. Uncomplicated myocardial infarction
b. Dyspnea without hypoxemia
c. Cardiac and respiratory arrest
d. Angina

Q27. Which of the following conditions is commonly associated with splinter hemorrhages?

a. Infective endocarditis
b. Pulmonary hypertension
c. Venous insufficiency
d. Aortic stenosis

Q28. Which of the following conditions is a cause of widened pulse pressure?

a. Cardiac tamponade
b. Aortic stenosis
c. Mitral regurgitation
d. Aortic regurgitation

Q29. Which of the following is a common ECG finding in patients with mitral regurgitation?

a. Prolonged PR interval
b. Shortened QT interval
c. Left atrial enlargement
d. ST-segment depression in the precordial leads

Q30. How does the act of squatting affect the venous return to the heart?

a. It increases preload
b. It decreases preload
c. It has no effect on preload
d. It can either increase or decrease preload depending on the individual

Q31. A 45 year-old man, presented to the outpatient clinic complaining of bilateral lower edema and increase abdominal contour within the previous 3 months. Examination of the patient revealed congested neck veins, this finding indicate ............?

a. Decreased venous return
b. Increased central venous pressure (CVP)
c. Chronic left side heart failure
d. Acute left side heart failure

Q32. Which of the following is a cause of orthopnoea in patients with left ventricular failure?

a. Increased venous return when lying flat
b. Decreased venous return when lying flat
c. Right ventricular dysfunction
d. Aortic stenosis

Q33. Collapsing pulse is commonly associated with which of the following conditions?

a. Aortic stenosis
b. Atrial fibrillation
c. Cardiac tamponade
d. Aortic regurgitation , whater hummur Pulse

Q34. Which of the following is a potential cause of secondary aortic regurgitation?

a. Aortic dissection
b. Atherosclerosis
c. Pulmonary embolism
d. Hypertension

Q35. A 30-year-old male is noted to have a high-pitched pansystolic murmur at the apex that radiates to the axilla. There is a mid-diastolic rumble at the apex. What is the most likely cause of this murmur?

a. Tricuspid regurgitation
b. Mitral stenosis
c. Ventricular septal defect
d. Mitral regurgitation

Q36. Which of the following is a characteristic feature of Janeway lesions?

a. Painful nodules on the palms, soles and distal phalanges
b. Painless flat lesions on the palms, soles and distal phalanges
c. Itchy, raised lesions on the fingertips
d. Painless raised lesions on the palms of the hand

Q37. Which of the following is a common cause of ventricular septal defect?

a. Congenital heart defect
b. Rheumatic fever
c. Endocarditis

d. Aortic stenosis

Q38. What is the underlying cause of the fourth heart sound?

a. Atrial contraction
b. Rapid ventricular filling
c. Blood flow across a stenotic valve
d. Relaxation of the ventricles

Q39. Which of the following is a common symptom in patients with right ventricular failure?

a. Palpitations
b. Dyspnoea on exertion
c. Peripheral oedema
d. Haemoptysis

Q40. Which of the following treatments is typically recommended for severe symptomatic aortic regurgitation?

a. Pharmacologic management
b. Balloon valvuloplasty
c. Aortic valve replacement
d. Coronary artery bypass grafting

Q41. Which of the following conditions is a cause of pulsus paradoxus?

a. Aortic stenosis
b. Atrial fibrillation → Pulsus defecit
c. Aortic regurgitation
d. Cardiac tamponade

Q42. Which of the following is a potential cause of carotid bruits?

a. Carotid artery stenosis
b. Aortic regurgitation
c. Aortic aneurysm
d. Carotid artery dissection

Q43. ECG identified by the PR interval tends to become longer with every succeeding ECG complex until there is a P wave not followed by a QRS is observed in

a. Third-Degree Atrioventricular Block
b. Second-Degree Atrioventricular Block, Type II

c. Second-Degree Atrioventricular Block, Type I.
d. First-Degree Atrioventricular Block, Type II.

Q44. What is the primary mechanism by which atherosclerosis leads to coronary artery disease (CAD)?
a. Infection of the heart muscle
b. Accumulation of plaques in the coronary arteries
c. Inflammation of the heart valves
d. Overproduction of red blood cells

Q45. Which of the following is the most common cause of pneumomediastinum in older children and teenagers ?
a. Dental extractions
b. Adenotonsillectomy
c. Tuberculosis
d. Acute asthma

Q46. Ali is 2year-old boy attends the Emergency room by acute onset of cough, dyspnoea and stridor while he was playing with his toys. Which of the following is immediate management should be done?
a. Thoracotomy.
b. Tracheostomy.
c. Bronchoscopy.
d. Mechanical Ventilation.

Q47. What would be an appropriate first-line treatment for the management of cyanosis as a result of heart disease?
a. Indomethacin administration
b. Prostaglandin E1 infusion
c. Nitric oxide
d. Surfactant

Q48. A child presents with recurrent sinusitis and recurrent chest infections. Chest X-ray reveals dextrocardia and situs inversus. Which of the following isThe most likely diagnosis ?
a. Kartagener's syndrome
b. Good-pasture's syndrome
c. Ehlers–Danlos syndrome

d. William Campbell syndrome

Q49. Which of the following is used to reduce friction between the layers of membranes surrounding the heart?

a. Synovial fluid
b. Pleural fluid
c. Pericardial fluid
d. Capillary endothelium

Q50. The apex of the heart is normally pointed

a. to the left of the midline
b. to the right of the midline
c. is different for males and females
d. posteriorly

Q51. Which of the following is referred to as the pacemaker of the heart?

a. Sinoatrial node
b. Purkinje fibers
c. Atrioventricular node
d. Endocardium

Q52. Why the pulmonary artery blood pressure (BP) is lower than the systemic arterial BP ?

a. The output of the right ventricle is less than that of the left ventricle.
b. The left ventricular wall is thicker than that of the right ventricle.
c. The aorta is more elastic than the pulmonary artery.
d. The pulmonary resistance is less than the systemic resistance.

Q53. What is the cardiac ejection fraction?

a. It is the percentage ratio of cardiac work to total energy expenditure.
b. It is the cardiac output per square meter body surface area.
c. It is the percentage of blood leaving the heart in each cardiac cycle.
d. It is a premature contraction of the heart occurring during diastole.

Q54. A 75-year-old woman with a history of chronic congestive heart failure undergoes cardiac catheterization to determine her extent of cardiac dysfunction. During estimation of her systolic function, in what phase of the cardiac cycle should her left ventricular volume unchanged?

a. Rapid filling
b. Isovolumetric contraction
c. Ventricular ejection
d. Atrial systole

Q55. Which beta-2 agonist provides 24-hour bronchodilation, allowing once-daily dosing?
a. Albuterol
b. Salmeterol
c. Terbutaline
d. Indacaterol

Q56. Which of the following bronchodilators is a long-acting beta-2 agonist and commonly used alongside ICS in asthma or chronic obstructive pulmonary disease (COPD)?
a. Albuterol
b. Salmeterol
c. Terbutaline
d. Salbutamol

Q57. The half-life of theophylline is shortened by which of the following medications, due to increased hepatic enzyme induction?
a. Erythromycin
b. Phenobarbitone
c. Ciprofloxacin
d. Cimetidine

Q58. Which of the followings lung tumor has a tendency to undergo necrosis and cavitations?
a. Adenocarcinoma
b. Bronchioalveolar carcinoma
c. Carcinoid
d. Squamous cell carcinoma

Q59. What is Gross feature of benign hypertension on kidney?
a. Contracted granular kidney
b. Enlarged kidney with petechial hemorrhage
c. Posollated kidney
d. Enlarged Vacuolated kidney

Q60. Which of the followings inhalation of particle is an important factor in development of Mesothelioma?

a. Bird dust
b. Silica
c. Asbestos
d. Coal dust

## Short Answer Question (SAQ)

**Write in the following 8 SAQs: Total 8 marks (1 mark for each question)**

Q1. Enumerate 4 diagnostic tests of interstitial lung diseases

1. Pulmonary Function test
2. CT
3. BAL
4. Surgical Lung biopsy

Q2. Enumerate 4 features of exudative pleural effusion

1.
2. pleural fluid protein to serum protein is > 0.5.
3. pleural fluid LDH to serum LDH is > 0.6.
4. 3. Cloudy or Turbid Appearance

low PH and glucose

Q3. Imagine you are a blood cell in the superior vena cava (SVC). Describe your journey as you move through the cardiovascular system until you get to the aortic arch?

Q4. List four indications for termination of Stress ECG test

Drop in Systolic Blood Pressure

Moderate to Severe Angina

Central Nervous System Symptoms

Symptoms like ataxia (stumbling), dizziness, or near-syncope

Sustained Ventricular Tachycardia or Serious Arrhythmias

Q5. Describe the internal features of right atrium.

smooth posterior part (sinus venarum) and a rough anterior part (atrium proper), separated by a ridge called the crista terminalis.

Q6. List four factors affecting pulmonary blood pressure and flow?

1. Hypoxic Pulmonary Vasoconstriction (HPV)
2. Lung Volume and Alveolar Pressure
3. Gravity and Body Position
4. Cardiac Output and Recruitment

Q7. List four therapeutic indications to morphine in cardiothoracic emergency?

1. Acute Myocardial Infarction
2. Aortic Dissection
3. Acute Pulmonary Edema
4. Post-Operative Cardiothoracic Pain

Q8. List etiology of myocardial infarction (four)?

1. Atherosclerotic Plaque Rupture

Supply-Demand Mismatch

Coronary Artery Spasm



Coronary Artery Embolism

2. ____________________
3. ____________________
4. ____________________

## Extended Matched Question (EMQ)

Write in the following 2 EMQs: Total 4 marks (Two marks for each question)

Each group of extended matching questions (EMQs) consists of a list of problems followed by a lettered set of options. For each numbered problem, select the lettered option that most closely diagnoses the problem. You can use the letter option once or not at all. Only put the letter corresponding to your answer.

Q1. Each group of extended matching questions consists of a list of numbered problems (diseases) followed by lettered options (Most presenting symptoms). For each numbered disease, select the lettered most presenting symptoms that most closely diagnose the disease. Total two marks (0.5 mark for each presentation)

- **Problems:**
  1. Pulmonary tuberculosis (A)
  2. Acute massive pulmonary embolism (E)
  3. Lung cancer (F)
  4. Bronchiectasis (G)
- **Options:**

1 (A) Fever and sweating especially at night

(B) Progressive dyspnea

(C) Bilateral lower limb edema

(D) Cyanosis

2 (E) Acute dyspnea and acute chest pain

3 (F) Hemoptysis, chronic chest pain and weight loss

4 (G) Expectoration of large amount of sputum

Q2: For each of the following patients' scenario (problem), select the most likely diagnosis (option). **Total two marks (0.5 mark for each scenario)**

- **Problems:**

B) Aortic regurgitation

1. A 70-year-old male has a decrescendo, early diastolic murmur heard best at the left lower sternal border. The murmur increases with expiration. What is the most likely cause of this murmur? ( )
2. A 65-year-old male is noted to have a harsh ejection systolic murmur heard best at the second right intercostal space with radiation to the neck. The murmur decreases with the Valsalva maneuver and increases with squatting. What is the most likely cause of this murmur? C) Aortic stenosis ( )
3. A 24-year-old male has a harsh ejection systolic murmur heard best at the left lower sternal border. The murmur increases with Valsalva maneuver and decreases with squatting. What is the most likely cause of the murmur? (E) Hypertrophic cardiomyopathy
4. A 30-year-old male is noted to have a high-pitched pansystolic murmur at the apex that radiates to the axilla. There is a mid-diastolic rumble at the apex. What is the most likely cause of this murmur? A) Mitral regurgitation ( )

- **Options:**
  (A) Mitral regurgitation
  (B) Aortic regurgitation
  (C) Aortic stenosis
  (D) Pulmonary stenosis
  (E) Hypertrophic cardiomyopathy
  (F) Ventricular septal rupture

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## Modified Essay Questions (MEQs)

**Write in the following 2 MEQs: Total 4 marks (Two marks for each question)**

Q1. Write about the causes of mediastinal masses by location.

Anterior Mediastinum: Thymoma , Teratoma

Middle Mediastinum: Congenital Cysts

Posterior Mediastinum as

Neurogenic Tumors: These are the most common cause of posterior masses.

Schwannomas and Neurofibromas

Q2. Mention investigations of bronchogenic carcinoma.

1. Laboratory Investigations

CBC, Liver Function Tests (LFTs), and Kidney Function Tests (KFTs

2. Imaging (The First Step) as

Chest X-ray + CT scan, PET

Brain MRI

3. Tissue & Cytological Diagnosis (The Confirmatory Step)

## Case Solve Problems (CSP)

**Write in the following (7 Cases): Total 14 marks (Two marks for each case)**

Case (1): An 18-year-old female, a non-smoker, university student, with a known case of bronchial asthma on long-acting beta-2 agonist and low-dose inhaled corticosteroids (ICS). She presents with sudden marked shortness of breath and impaired conscious level. She was cyanotic and unable to speak one sentence in one breath, pulse: 132, b/m, RR: 32 C/m, blood pressure: 94/58 mmHg, lung auscultation revealed markedly diminished breath sounds with distant polyphonic expiratory wheezes through the entire chest. ABG reveals O2 sat: 84, PaO2: 49 mmHg, PaCO2:76 mmHg, HCO3:21, pH: 7.26. She received all the necessary medications in the ER, but the condition did not improve, and she was transferred to the intensive care unite (ICU).

1. What is the diagnosis of this case? **(1 mark)**

Life-threatening Asthma Exacerbation) with Type II (Hypercapnic) Respiratory Failure.

2. Enumerate two features of this clinical presentation. **(0.5 mark)**

Hypercapnia (PaCO, 76 mmHg

The "Silent Chest"

Hemodynamic instability: Hypotension (BP 94/58)

3. What is the type of respiratory failure in this case? **(0.5 mark)**.

**Case (2)**: A 66-year-old male, smoker with a 37-pack-year history. He is a farmer, earning pigeon, and a resident near a cement factory. He presents with a dry cough and gradual worsening of breathlessness over the last 3 years. He has clubbing, and his chest auscultation showed fine, late inspiratory, bi-basal crackles. He has congested pulsating neck veins, lower limb edema, enlarged tender liver, and accentuated pulmonary component of the 2nd heart sound. High-resolution computed tomography (CT) scan showed reticular abnormality, ground-glass opacity, centrilobular nodules, and traction bronchiectasis. His pulmonary function test revealed; FEV1/FVC ratio:90%, FEV1:82%, FVC:51%, Total lung capacity (TLC): 58%, and DLCO:49%

1. What is the diagnosis of this case? **(1 mark)**

The diagnosis is Hypersensitivity Pneumonitis (HP), likely the chronic/fibrotic form, which

has led to Interstitial Lung Disease (ILD).

2. Enumerate two risk factors for this disease in the present case **(0.5 mark)**.

Avian exposure (Pigeon breeding)

Environmental/Occupational exposure (Cement factory)

3. Name the presence of congested pulsating neck veins, lower limb edema, enlarged tender liver, and accentuated pulmonary component of the 2nd heart sound in this case **(0.5 mark)**.

The presence of congested pulsating neck veins, lower limb edema, an enlarged tender liver (congestive hepatomegaly), and an accentuated $P_2$ (pulmonary component of the second heart sound) is called Cor Pulmonale.

**Case (3):** A 59-year-old male presents in the emergency room (ER) of your hospital with crushing chest pain that radiates to the jaw or shoulder for the past 30 minutes. He also feels shortness of breath and nauseous. O2 is 96% and you carry out an ECG and note that there is ST elevation in leads II, III, and AVF.

a. What part of the heart is likely to be affected by this myocardial infarction, and which coronary artery is most likely to be implicated?

Part of the heart affected: This is an Inferior Wall Myocardial Infarction (IWMI).

Right Coronary Artery (RCA)

b. Which should be involved in your acute management of this ST segment elevation myocardial infarction (STEMI)?

Morphine Oxygen

Aspirin Nitrates

**Case (4):** A 45-year-old male recently had a coronary artery bypass graft (CABG) after suffering a myocardial infarction 2 weeks ago. He now has a sharp pain in his chest, which is radiating to his left shoulder, and is worse when he takes a deep breath in. He says the pain is lessened when he is sitting forward compared to when he is lying flat. On examination, his vital signs are normal, and there are no murmurs on auscultation. His troponin blood test comes back normal.

a. What is the most likely diagnosis?

Acute Pericarditis.

b. Which should be involved in your management of this case?

NSAIDs (Non-Steroidal Anti-inflammatory Drugs

Colchicine

Monitoring

Activity Modification

**Case (5):** A 54-year-old woman presents with shortness of breath on exertion, fatigue, and palpitations. On examination, her jugular venous pressure is elevated, and a pansystolic, high-pitched "whistling" murmur is heard. The murmur radiates to the left axilla.

a. What is the likely cause of the murmur?

The likely cause is Mitral Regurgitation (MR

b. What is the most appropriate diagnostic investigation?

is Transthoracic Echocardiography (TTE).

**Case (6):** A 2-month-old girl presents to the emergency department with respiratory distress. Her mother reports that the infant had been well until about 2 weeks ago, when she first noticed the infant had rapid breathing. The infant had previously been fed well, but now becomes sweaty and irritable when taking a bottle. The mother is concerned that the infant is losing weight. Physical examination reveals a HR of 140/min, BP 80/45 mmHg, RR 60/min, and temperature 36.8 °C. There is a prominence of the left precordium and a palpable sternal lift. She has a blowing holosystolic murmur that is best heard along the left sternal border. SpO2 is 95%. Her chest x-ray shows cardiomegaly and increased pulmonary vascular markings.

a. What is the most likely diagnosis?

Large Ventricular Septal Defect (VSD).

b. What is the management?

1. Medical Management (Stabilization)

Diuretics

Afterload Reducers: (e.g., ACE inhibitors like Captopril

Nutritional Support

2. Surgical Management (Definitive Treatment

Surgical Repair

Case (7): A 62-year-old male presents to the outpatient clinic with complaints of increasing fatigue, swelling in his legs, and abdominal discomfort for the past month. He has a history of long-standing hypertension and chronic obstructive pulmonary disease (COPD). On examination, his blood pressure is 130/80 mmHg, heart rate is 92 bpm, and oxygen saturation is 94% on room air. You observe jugular venous distension, hepatomegaly, and bilateral pitting edema in the lower limbs. Lung auscultation reveals clear breath sounds bilaterally.

a. What is the most likely diagnosis? **(0.5 mark)**

Cor Pulmonale (Right-sided Heart Failure

b. Which is the *most likely cause* of his current condition? **(0.5 mark)**

Chronic Obstructive Pulmonary Disease (COPD

c. What are the treatment options? **(1 mark)**

1. Management of the Underlying Cause (Most Critical)

Oxygen Therapy: Long-term oxygen

Bronchodilators and Inhaled Corticosteroids

Smoking Cessation

2. Symptomatic Management of Heart Failure

Diuretics

Salt and Fluid Restriction

*Good Luck*

Case (7): A 62-year-old male presents to the outpatient clinic with complaints of increasing fatigue, swelling in his legs, and abdominal discomfort for the past month. He has a history of long-standing hypertension and chronic obstructive pulmonary disease (COPD). On examination, his blood pressure is 130/80 mmHg, heart rate is 92 bpm, and oxygen saturation is 94% on room air. You observe jugular venous distension, hepatomegaly, and bilateral pitting edema in the lower limbs. Lung auscultation reveals clear breath sounds bilaterally.

a. What is the most likely diagnosis? **(0.5 mark)**

b. Which is the *most likely cause* of his current condition? **(0.5 mark)**

c. What are the treatment options? **(1 mark)**

*Good Luck*